

The Ja Model: Methodology and Empirical Demonstration of the MUja Framework (Damped Oscillator Reconstruction and GRACE-FO 1 Satellite Drag Analysis)

Author: Jalal Khawaldeh

Director, NourScene Research Initiative

ORCID: 0009-0003-7872-1967

Email: jalal@nourscene.pro | Correspondence: jalal.khawaldh@yahoo.com

DOI: [10.5281/zenodo.20551386](https://doi.org/10.5281/zenodo.20551386)

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Abstract

This paper presents the complete methodology and empirical evaluation of the MUja framework, a dimensionless comparative index for dissipative processes defined as:

$$MUja = (F \cdot d)/(m \cdot E) \cdot m_0 \cdot (\Delta S/k_B)$$

where F is force, d displacement, m mass, E energy, ΔS entropy change, m_0 a reference mass, and k_B Boltzmann's constant. The framework replaces conventional time with a cycle count N , where N counts periods of a reference periodic process.

The methodology is demonstrated on two physically distinct dissipative systems:

1. **Damped harmonic oscillator** – reconstructed from published fitted parameters (Sebastião 2014). The oscillator parameters are: $m = 0.10219$ kg, $K = 10.392$ N·m⁻¹, $\lambda = 0.13878$ s⁻¹.

The amplitude decay $A(t) = (Zt_0 - Z\infty)\exp(-\lambda t) + Z\infty$ is used to compute dissipated energy per cycle $\Delta E_{diss} = \pi b \omega A^2$ and $MUja = (m_0/m) \cdot (\pi b \omega A^2)/(k_B T)$. The reconstructed MUja evolution follows $MUja(N) \propto \exp(-0.173N)$, with a dimensionless decay constant of 0.173 cycle⁻¹. Note: these numerical values are derived from the fitted model and are illustrative, not direct experimental measurements.

2. **GRACE-FO 1 satellite (NORAD 43476) under atmospheric drag** – using 2680 orbital records from 30 November 2023 to 31 May 2026 (912.42 days). The semi-major axis altitude decreased from 483.379 km to 451.615 km (total loss 31.764 km), corresponding to a total orbital energy loss $\Delta E_{total} = 5.836 \times 10^7$ J. Over 14,034 orbital revolutions, the mean dissipated energy per orbit is 4.158×10^3 J, yielding a mean MUja of 6.972×10^{20} . The full MUja distribution ($n = 2608$ valid adjacent intervals) gives: mean = 7.126×10^{20} , median = 6.421×10^{20} , standard deviation = 3.558×10^{20} , 95% bootstrap CI = [7.00×10^{20} , 7.29×10^{20}], skewness = 2.425, indicating a right-tailed distribution consistent with episodic atmospheric disturbances (e.g., geomagnetic storms).

The oscillator analysis is a proof-of-concept consistency test based on a fitted model. The satellite analysis uses real orbital observations and demonstrates operational feasibility. The framework does not claim new physical laws; it provides a dimensionless, cycle-based language for comparing dissipative processes across vastly different scales. Limitations include the reliance on fitted parameters for the oscillator, the assumed constant thermospheric temperature ($T = 1000$ K) for the satellite, and the exclusion of conservative systems, massless particles, and relativistic effects.

Keywords: MUja, dimensionless index, dissipative processes, cycle count, damped oscillator, satellite drag, GRACE-FO 1, entropy, reparametrisation, physics, applied mathematics, aerospace engineering, non-equilibrium thermodynamics, space weather, metrology.

Introduction

The description of dissipative (irreversible) processes in physics conventionally relies on time as an independent parameter. However, time itself is operationally defined through periodic motion – rotations, oscillations, atomic transitions. This circularity suggests that motion cycles might serve as a more primitive observable.

The Ja Model (Khawaldeh, Jalal Y. A. 2026e) proposes a descriptive framework where the conventional time parameter t is replaced by a cycle count $N = t/\tau_0$, with τ_0 the period of a chosen reference cycle. The central quantity is the dimensionless comparative index:

$$\mathbf{MUja} = (\mathbf{F} \cdot \mathbf{d}) / (\mathbf{m} \cdot \mathbf{E}) \cdot \mathbf{m}_0 \cdot (\Delta \mathbf{S} / \mathbf{k}_B)$$

where F is force, d displacement, m mass, E energy, ΔS entropy change, m_0 a reference mass, and k_B Boltzmann's constant.

For dissipative processes where the dissipated energy ΔE_{diss} equals $F \cdot d$ (e.g., viscous damping, atmospheric drag), and where the dissipation occurs isothermally, the entropy change of the environment is $\Delta S_{\text{env}} = \Delta E_{\text{diss}} / T$. Consequently, $\Delta S/k_B = \Delta E_{\text{diss}}/(k_B T)$. The expression then simplifies to:

$$\mathbf{MUja} = (\mathbf{m}_0/\mathbf{m}) \cdot (\Delta \mathbf{E}_{\text{diss}}) / (\mathbf{k}_B \mathbf{T})$$

with T the absolute temperature of the environment. The general form with $\Delta S/k_B$ is retained for future extensions to non-isothermal or information-theoretic contexts.

The Ja Model is a reparametrisation of non-relativistic physics, not a new physical law. It does not eliminate time but reinterprets it as a derived count. The present paper provides the complete methodology and empirical demonstration of the MUja framework on two radically different dissipative systems: a laboratory-scale damped harmonic oscillator (reconstructed from published data) and a space-based satellite (GRACE-FO 1) experiencing atmospheric drag. The objective is to test whether MUja can be computed consistently, remains finite and interpretable, and allows a common cycle-based description across scales.

This paper is self-contained and does not require prior reading of the theoretical Ja Model paper. All necessary definitions, equations, data sources, and computational steps are provided herein.

1. Methodology

1.1 Objective of the Empirical Study

The present study constitutes the first empirical evaluation of the MUJa framework using a real experimentally characterized dissipative physical system.

The objective of this investigation is not to establish a new law of physics, nor to replace existing dynamical theories. Instead, the purpose is to determine whether the MUJa descriptor can function as a meaningful dimensionless measure of dissipative motion when applied to experimentally observed systems.

The central research question is:

Can a single dimensionless quantity provide a consistent characterization of dissipative motion independently of the physical scale of the system?

To address this question, the damped harmonic oscillator was selected as the first test system because it is one of the most extensively studied dissipative systems in classical physics and possesses a well-established theoretical and experimental description.

The present analysis should be regarded as a proof-of-concept evaluation of the MUJa framework. Its objective is not to validate a new physical law, but to determine whether MUJa behaves consistently when applied to a real dissipative system whose dynamics are already well understood.

1.2 Central Hypothesis

The empirical analysis is based on the following operational hypothesis:

Dissipative systems may possess a common dimensionless structure that can be quantified through the MUJa descriptor.

More specifically:

If MUJa captures a physically meaningful aspect of dissipative motion, its evolution should systematically reflect changes in dissipative energy loss within a dynamical system.

For the present study, the hypothesis is formulated as:

H₁

For a damped harmonic oscillator, as dissipative energy loss per cycle decreases during the evolution of the system,

$\Delta E_{\text{diss}}(N) \downarrow$

the corresponding value of

MUja(N)

should also decrease in a consistent, reproducible, and physically interpretable manner.

Within the reconstructed oscillator model, this implies that the evolution of MUja should remain correlated with the evolution of dissipative energy loss throughout the observed cycle sequence.

Failure to observe such behavior would indicate that MUja does not provide a useful descriptor of dissipative evolution for the system under consideration.

The present hypothesis does not claim that MUja constitutes a new physical law. Rather, it tests whether MUja can serve as a stable and operationally meaningful dimensionless descriptor of dissipative processes.

1.3 Choice of Experimental System

The damped harmonic oscillator was selected for four primary reasons.

Reason 1: Physical Simplicity

The oscillator is governed by the classical equation of motion:

$$m \frac{d^2x}{dt^2} + b \frac{dx}{dt} + Kx = 0$$

where:

- m is the oscillating mass
- b is the viscous damping coefficient
- K is the spring constant

The physical interpretation of each parameter is well established and experimentally measurable.

Reason 2: Controlled Dissipation

The rate of energy loss is analytically known and experimentally measurable.

This allows direct estimation of:

ΔE_{diss}

without introducing speculative assumptions or model-dependent corrections.

Reason 3: Availability of Experimental Data

The work of Sebastião (2014) provides experimentally fitted parameters derived from a real laboratory spring-mass system.

The published study reports:

- mass
- spring constant
- damping coefficient
- amplitude decay law
- resonance characteristics

These quantities are sufficient to reconstruct the dissipative dynamics required for the MUja analysis.

Reason 4: Reproducibility

The experiment is sufficiently simple to be independently replicated.

Consequently, any conclusions obtained from the MUja analysis can be verified through future experimental studies.

2. Source Study

2.1 Overview of Sebastião (2014)

The experimental foundation of the present analysis is the spring-mass oscillator investigated by:

Sebastião, P. J. (2014).

The Art of Model Fitting to Experimental Results.

European Journal of Physics, 35, 015017.

The original purpose of that work was not the evaluation of MUja.

Instead, the study demonstrated methods for fitting physical models to experimental observations.

Among the examples presented is a damped spring-mass oscillator characterized through:

- static spring elongation measurements
- amplitude decay measurements
- resonance measurements
- phase-response measurements

The supplementary material provides fitted parameter values obtained from the experimental observations.

2.2 Parameters Available from the Experimental Study

The global fit reported in the supplementary material yields:

Oscillating Mass

$m = 0.10219 \text{ kg}$

Spring Constant

$K = 10.392 \text{ N}\cdot\text{m}^{-1}$

Damping Constant

$$\lambda = 0.14291 \text{ s}^{-1}$$

Initial Oscillation Amplitude

$$A_0 = 0.1045 \text{ m}$$

Asymptotic Offset

$$Z_\infty = -0.00246 \text{ m}$$

These parameters provide all quantities required to reconstruct the oscillator dynamics.

3. Reconstruction of the Experimental Dynamics

3.1 Data Availability and Reconstruction Strategy

The supplementary material provides fitted physical parameters and fitted decay functions, but does not provide the original raw amplitude table in the form:

(t, A)

Therefore, the present analysis reconstructs the oscillator dynamics from the published fitted parameters.

The analysis uses the published fitted parameters, not the raw experimental amplitude measurements. Consequently, the numerical values of MUJa reported herein should be interpreted as illustrative estimates of the expected behavior.

However, the functional form of the oscillator decay is determined directly from experimentally fitted parameters and therefore remains physically meaningful and robust.

3.2 Amplitude Evolution

The amplitude decay law reported in the supplementary data is:

$$A(t) = (Z_{t_0} - Z_\infty) e^{(-\lambda t)} + Z_\infty$$

with:

$$Z_{t_0} = 0.10319 \text{ m}$$

$$Z_{\infty} = -0.0032395 \text{ m}$$

$$\lambda = 0.13878 \text{ s}^{-1}$$

This equation describes the progressive reduction of oscillation amplitude due to dissipative losses.

3.3 Oscillation Frequency

The natural angular frequency is:

$$\omega_0 = \sqrt{K/m}$$

Substituting the experimental values:

$$\omega_0 = \sqrt{10.392 / 0.10219}$$

$$\omega_0 \approx 10.083 \text{ rad}\cdot\text{s}^{-1}$$

The corresponding oscillation frequency is:

$$f = \omega_0 / (2\pi)$$

$$f \approx 1.605 \text{ Hz}$$

The oscillation period becomes:

$$T_{osc} = 1/f$$

$$T_{osc} \approx 0.623 \text{ s}$$

3.4 Viscous Damping Coefficient

For a damped oscillator:

$$\lambda = b / (2m)$$

Therefore:

$$b = 2m\lambda$$

Substituting the experimental values:

$$b = 2 \times 0.10219 \times 0.13878$$

$$b \approx 0.02836 \text{ kg}\cdot\text{s}^{-1}$$

4. MUja Evaluation Procedure

4.1 Energy Dissipated per Oscillation Cycle

For a viscously damped oscillator, the energy dissipated during one oscillation cycle is:

$$\Delta E_{\text{diss}} = \pi b \omega A^2$$

where:

- b is the damping coefficient
- ω is the angular frequency
- A is the instantaneous oscillation amplitude

This expression represents the mechanical energy converted into dissipative losses during one oscillation cycle.

4.2 Temperature Assumption

The original experiment does not explicitly report laboratory temperature.

For consistency, room temperature is assumed to be:

$$T = 293 \text{ K}$$

A variation of $\pm 5 \text{ K}$ changes the absolute value of MUja by less than approximately 2%, because:

$$\text{MUja} \propto 1/T$$

Such a variation does not affect the qualitative behavior of MUja or the conclusions of the present study.

4.3 Simplified MUja Form

For the present analysis:

$$\text{MUja} = (m_o/m) \times (\Delta E_{\text{diss}}/(k_B T))$$

where:

$$m_o = 1 \text{ kg}$$

$$k_B = 1.380649 \times 10^{-23} \text{ J}\cdot\text{K}^{-1}$$

$$T = 293 \text{ K}$$

Substituting the dissipated energy expression:

$$\text{MUja} = (m_o/m) \times (\pi b \omega A^2)/(k_B T)$$

5. Predicted MUja Behavior

The amplitude decay law implies:

$$A(t) \propto e^{(-\lambda t)}$$

Therefore:

$$A^2(t) \propto e^{(-2\lambda t)}$$

Since the present formulation defines MUja in terms of the dissipated energy per cycle, and since the dissipated energy of a viscously damped oscillator is proportional to A^2 , it follows that:

$$\text{MUja}(t) \propto e^{(-2\lambda t)}$$

This relationship should not be interpreted as an independent physical prediction. Rather, it is an internal consistency expectation arising from the adopted MUja definition and the experimentally fitted amplitude-decay law.

Thus, the purpose of the oscillator analysis is to verify whether the reconstructed MUja evolution remains mathematically well-defined, monotonic, and consistent with the known dissipative behavior of the system.

5.1 Cycle-Based Representation

Within the Ja framework, the natural evolution parameter is the cycle count:

$$N$$

Using:

$$t = \tau_o N$$

where the reference period is chosen as the oscillation period:

$$\tau_0 = T_{osc} = 0.623 \text{ s}$$

the decay law becomes:

$$MU_{ja}(N) \propto \exp(-2\lambda\tau_0 N)$$

Substituting the experimental values:

$$2\lambda\tau_0 = 2 \times 0.13878 \times 0.623$$

$$2\lambda\tau_0 \approx 0.173$$

Therefore:

$$MU_{ja}(N) \propto \exp(-0.173N)$$

The quantity:

$$0.173 \text{ cycle}^{-1}$$

represents the dimensionless decay constant of the oscillator within the Ja representation.

6. Expected Scientific Outcome

The objective of the present study is not to establish a new physical law.

Rather, the objective is to determine whether the MUja framework provides a mathematically consistent and operationally useful dimensionless representation of dissipative processes.

A successful outcome would demonstrate that:

1. **MUja can be computed directly from experimentally derived quantities without introducing additional fitted parameters.**
2. **MUja remains finite, well-defined, and physically interpretable throughout the observed evolution of the system.**
3. **The cycle-based representation and the conventional time-based representation remain mutually consistent.**

4. **The computed MUja evolution remains consistent with the independently established dissipative behavior of the system.**
5. **The framework can be applied reproducibly to real experimental data using a transparent calculation procedure.**

The present study is not intended to validate the universality of MUja.

Rather, it constitutes an initial proof-of-concept investigation designed to determine whether the framework can be applied consistently to a real experimentally characterized dissipative system.

Any broader claims regarding generality or cross-domain applicability require additional analyses involving independent dissipative systems beyond the oscillator considered here.

7. Results: Reconstructed MUja Evolution for a Damped Harmonic Oscillator

7.1 Purpose of the Analysis

The objective of the present analysis is to determine whether the MUja descriptor evolves consistently with the experimentally observed dissipation of mechanical energy in a damped harmonic oscillator.

The present investigation constitutes the first empirical proof-of-concept evaluation of the MUja framework using a real experimentally characterized dissipative system.

Because the original supplementary material provides fitted physical parameters rather than a complete raw amplitude dataset, the oscillator dynamics were reconstructed from the published decay model. Consequently, the numerical values reported herein should be interpreted as reconstructed estimates derived from experimentally fitted parameters rather than as a direct reanalysis of the original laboratory measurements.

The analysis therefore uses the published fitted parameters rather than the original raw amplitude table in the form:

(t, A)

Accordingly, the numerical MUja values are illustrative of the expected behavior of the system. However, the functional form of the MUja evolution follows directly from experimentally fitted parameters and therefore remains physically meaningful and mathematically well-defined.

The purpose of this section is not to establish a new physical law. Rather, it is to determine whether MUja behaves consistently with the dissipative evolution of a well-characterized physical system and whether the framework can be implemented transparently using experimentally derived quantities.

7.2 Reconstruction of Oscillator Dynamics

The published amplitude decay law is:

$$\mathbf{A(t) = (Zt_o - Z\infty) \exp(-\lambda t) + Z\infty}$$

with:

$$\mathbf{Zt_o = 0.10319 \text{ m}}$$

$$\mathbf{Z\infty = -0.0032395 \text{ m}}$$

$$\mathbf{\lambda = 0.13878 \text{ s}^{-1}}$$

The oscillator parameters reported in the global fit are:

$$\mathbf{m = 0.10219 \text{ kg}}$$

$$\mathbf{K = 10.392 \text{ N}\cdot\text{m}^{-1}}$$

The natural angular frequency is:

$$\mathbf{\omega_o = \sqrt{K/m}}$$

Substituting the experimental values:

$$\mathbf{\omega_o = \sqrt{10.392/0.10219}}$$

$$\mathbf{\omega_o \approx 10.083 \text{ rad}\cdot\text{s}^{-1}}$$

The oscillation frequency is:

$$\mathbf{f = \omega_o/(2\pi)}$$

$$\mathbf{f \approx 1.605 \text{ Hz}}$$

The oscillation period is:

$$T_{osc} \approx 0.623 \text{ s}$$

The viscous damping coefficient is:

$$b = 2m\lambda$$

which yields:

$$b \approx 0.02836 \text{ kg}\cdot\text{s}^{-1}$$

Strictly speaking, the dissipative dynamics are governed by the damped angular frequency:

$$\omega_d = \sqrt{(\omega_o^2 - \lambda^2)}$$

However:

$$\lambda \ll \omega_o$$

and therefore:

$$\omega_d \approx \omega_o$$

The relative difference between the two frequencies is approximately:

$$(\omega_o - \omega_d)/\omega_o \approx 9.5 \times 10^{-5}$$

corresponding to less than 0.01%.

Consequently, the distinction between ω_o and ω_d introduces a negligible error in the computed MUja values and has no practical influence on the conclusions of the present analysis.

7.3 Dissipated Energy per Cycle

For a viscously damped oscillator, the mechanical energy dissipated during one oscillation cycle is:

$$\Delta E_{diss} = \pi b \omega A^2$$

where:

- b is the viscous damping coefficient,

- ω is the oscillation angular frequency,
- A is the instantaneous oscillation amplitude.

Since:

$$\Delta E_{\text{diss}} \propto A^2$$

the dissipated energy decreases continuously as the oscillation amplitude decays.

Using the reconstructed amplitude evolution, the dissipated energy was evaluated at successive oscillation cycles.

The resulting dissipation sequence provides the energetic basis for the subsequent MUja calculation.

7.4 MUja Computation

For the present analysis:

$$m_o = 1 \text{ kg}$$

$$T = 293 \text{ K}$$

$$k_B = 1.380649 \times 10^{-23} \text{ J}\cdot\text{K}^{-1}$$

The simplified MUja descriptor is evaluated as:

$$\text{MUja} = (m_o/m) \times (\Delta E_{\text{diss}}/(k_B T))$$

For the damped oscillator examined here, the energy term appearing in the general MUja definition is identified with the dissipated energy:

$$E = \Delta E_{\text{diss}}$$

Furthermore, the dissipative work performed during one oscillation cycle satisfies:

$$\mathbf{F} \cdot \mathbf{d} = \Delta E_{\text{diss}}$$

Therefore, the simplified MUja form employed in the present analysis is valid for this dissipative process.

Substituting the dissipated-energy expression yields:

$$\text{MUja} = (m_o/m) \times (\pi b \omega A^2)/(kB T)$$

Thus:

$$\text{MUja} \propto A^2$$

The numerical magnitude of MUja depends linearly on the selected reference mass:

m_o

A different normalization choice would therefore change the absolute numerical values of MUja.

However, such a rescaling would not alter:

- the decay law,
- relative comparisons,
- temporal evolution,
- qualitative behavior of the descriptor.

Consequently, the physically relevant information resides primarily in the evolution and relative scaling of MUja rather than in its absolute numerical magnitude.

7.5 Reconstructed MUja Evolution

Using the reconstructed oscillator dynamics, representative values of the dissipated energy and MUja were obtained.

Cycle N	Time (s)	Amplitude A (m)	ΔE_{diss} (J)	MUja
0	0.000	0.10319	9.57×10^{-3}	2.31×10^{19}
1	0.623	0.09437	8.00×10^{-3}	1.94×10^{19}
2	1.246	0.08629	6.69×10^{-3}	1.62×10^{19}
3	1.869	0.07887	5.59×10^{-3}	1.35×10^{19}
4	2.492	0.07207	4.67×10^{-3}	1.13×10^{19}
10	6.230	0.04079	1.50×10^{-3}	3.62×10^{18}

15	9.345	0.02576	5.96×10^{-4}	1.44×10^{18}
20	12.460	0.01564	2.20×10^{-4}	5.31×10^{17}
23	14.329	0.01132	1.15×10^{-4}	2.79×10^{17}

Note: All numerical values in this table are derived from the fitted amplitude-decay model (Sebastião 2014, supplementary material) and are illustrative. They do not represent direct experimental measurements.

The reconstructed results demonstrate a monotonic decrease in both dissipated energy and MUja throughout the dissipative evolution of the system.

No discontinuities, singularities, or nonphysical behavior appear within the reconstructed trajectory.

7.6 Expected Exponential Decay

The experimentally fitted amplitude envelope obeys:

$$A(t) \propto \exp(-\lambda t)$$

Therefore:

$$A^2(t) \propto \exp(-2\lambda t)$$

Since:

$$MUja \propto A^2$$

it follows that:

$$MUja(t) \propto \exp(-2\lambda t)$$

Here, λ is the same decay constant appearing in the experimentally fitted amplitude envelope:

$$A(t) = (Zt_0 - Z^\infty) \exp(-\lambda t) + Z^\infty$$

The factor of two arises naturally from the quadratic dependence of dissipated energy on oscillation amplitude.

Using the fitted value:

$$\lambda = 0.13878 \text{ s}^{-1}$$

the expected MUja decay constant becomes:

$$2\lambda = 0.27756 \text{ s}^{-1}$$

Consequently:

$$\text{MUja}(t) \propto \exp(-0.27756 t)$$

Because the present derivation follows directly from the experimentally fitted amplitude envelope, the resulting decay law should be interpreted as a model-consistency consequence rather than as an independently derived physical prediction.

This relationship nevertheless constitutes the principal theoretical expectation of the present analysis.

7.7 Cycle-Count Representation

Within the Ja framework, the natural evolution parameter is the cycle count:

N

Using:

$$t = T_{osc} N$$

with:

$$T_{osc} = 0.623 \text{ s}$$

the expected MUja evolution becomes:

$$\text{MUja}(N) \propto \exp(-2\lambda T_{osc} N)$$

Substituting the experimentally fitted values:

$$2\lambda T_{osc} = 2 \times 0.13878 \times 0.623$$

$$2\lambda T_{osc} \approx 0.173$$

Therefore:

$$\text{MUja}(N) \propto \exp(-0.173 N)$$

The quantity:

0.173 cycle⁻¹

represents the dimensionless decay constant associated with the reconstructed oscillator model within the cycle-based Ja representation.

Because the present MUja formulation is proportional to the dissipated energy per cycle and because the dissipated energy of the reconstructed oscillator is proportional to A^2 , the above expression follows directly from the experimentally fitted amplitude-decay law.

Accordingly, this relationship should be interpreted as the expected internal behavior of the reconstructed model rather than as an independent physical prediction.

7.8 Evaluation of the Operational Hypothesis

The operational hypothesis H_1 states:

As the cycle count N increases, the computed MUja evolution should remain consistent with the independently established dissipative behavior of the oscillator.

The reconstructed analysis satisfies this expectation.

Specifically:

1. The oscillation amplitude decreases exponentially according to the experimentally fitted decay law.
2. The dissipated energy per cycle decreases proportionally to A^2 .
3. The computed MUja values decrease consistently with dissipative energy loss.
4. The reconstructed MUja evolution follows the expected functional form derived from the fitted amplitude model.
5. No discontinuities, divergences, or nonphysical behavior appear during the evolution.

These observations demonstrate that the MUJa framework can be applied consistently to the reconstructed oscillator dynamics and remains mathematically well-defined throughout the observed evolution.

Because the analysis is based on the same fitted model used to derive the expected behavior, the results should be interpreted as a proof-of-concept consistency test rather than as an independent validation of the framework.

7.9 Limitations

The present analysis relies on published fitted parameters rather than the original raw amplitude measurements.

Consequently:

- the reported MUJa values should be regarded as reconstructed estimates,
- the results inherit the assumptions and limitations of the fitted decay model,
- the functional behavior of MUJa is more robust than its absolute numerical magnitude,
- the analysis does not provide an independent experimental test of the fitted oscillator dynamics.

The numerical magnitude of MUJa scales linearly with the selected reference mass:

m_0

Therefore, absolute MUJa values depend on the adopted normalization convention.

However, the reconstructed evolution law:

$$\mathbf{MUJa(N) \propto \exp(-0.173N)}$$

and all relative comparisons remain invariant under such rescaling.

Because the dataset is reconstructed from fitted functions, no experimental scatter is present.

In real measurements, uncertainties in amplitude estimation would propagate into the computed MUja values.

For example, an amplitude uncertainty of approximately:

$\pm 0.5\%$

would produce an MUja uncertainty of approximately:

$\pm 1\%$

because:

$\text{MUja} \propto A^2$

At late cycles, the fitted offset parameter:

Z_{∞}

becomes a progressively larger fraction of the reconstructed amplitude.

Consequently, the relative uncertainty of the reconstructed MUja values is expected to increase near the end of the decay sequence.

The uncertainty estimate above considers only amplitude uncertainty.

A complete uncertainty budget would additionally include uncertainties associated with:

- mass,
- damping coefficient,
- oscillation frequency,
- temperature,
- calibration procedures.

Future analyses based on raw experimental measurements should propagate all relevant uncertainties through the complete MUja calculation using standard error-propagation techniques or Monte Carlo methods.

7.10 Interim Conclusion

The damped harmonic oscillator constitutes the first system to which the MUJa framework has been applied using experimentally derived physical parameters.

The reconstructed analysis demonstrates that:

- MUJa can be computed directly from experimentally derived quantities,
- MUJa remains finite and mathematically well-defined throughout the observed evolution,
- the cycle-based representation remains consistent with the conventional time-based description,
- the computed MUJa evolution remains consistent with the dissipative behavior encoded in the fitted oscillator model.

Within the assumptions and limitations of the reconstructed analysis, the MUJa evolution follows:

$$\text{MUJa}(N) \propto \exp(-0.173N)$$

This result does not establish a new physical law, nor does it demonstrate the universality of the MUJa framework.

Rather, it shows that the framework can be implemented consistently in a real dissipative system using experimentally derived parameters and a transparent calculation procedure.

The present oscillator analysis therefore establishes computational feasibility, internal consistency, and operational transparency, but not independent empirical validation.

The broader question of whether MUJa provides useful cross-domain comparisons between fundamentally different dissipative systems remains open and requires independent analyses of additional systems beyond the oscillator considered here.

8. Robustness Analysis of the Oscillator MUja Results

8.1 Purpose of the Robustness Analysis

Before extending the MUja framework to additional dissipative systems, the oscillator results must be tested for internal robustness.

The purpose of this section is to determine whether the MUja behavior observed in the reconstructed damped-oscillator analysis remains stable with respect to:

1. the choice of reference mass m_o ,
2. the assumed environmental temperature T ,
3. amplitude measurement uncertainty,
4. alternative estimates of the damping parameter λ ,
5. the conversion from time-based decay to cycle-count decay.

This robustness analysis is necessary because the absolute numerical value of MUja depends on normalization choices, whereas the physically relevant question is whether its underlying evolution remains invariant.

The primary objective is therefore not to examine the absolute magnitude of MUja, but rather to determine whether its functional behavior remains stable under reasonable variations of the assumptions and parameters entering the calculation.

The analysis focuses on the stability of the descriptor's structure rather than on its specific numerical scale.

8.2 Offset-Corrected Amplitude Used for Robustness Testing

The published amplitude model includes an offset term:

$$A(t) = (Zt_o - Z^\infty) \exp(-\lambda t) + Z^\infty$$

where:

$$Z^\infty = -0.0032395 \text{ m}$$

This offset represents a fitted asymptotic correction and should not be interpreted as a physical residual oscillation amplitude.

Therefore, for robustness testing, the physically relevant oscillatory component is taken as the offset-corrected amplitude:

$$\mathbf{Acorr(t) = A(t) - Z_{\infty}}$$

Thus:

$$\mathbf{Acorr(t) = (Zt_0 - Z_{\infty}) \exp(-\lambda t)}$$

This correction ensures that the energy calculation is based exclusively on the oscillatory component relative to the fitted equilibrium level.

The reconstructed analysis therefore remains fully consistent with the experimentally reported decay model while avoiding artificial contributions arising from the fitted offset.

8.3 Baseline Robustness Dataset

Using the offset-corrected amplitude, the baseline robustness calculation adopts:

$$\mathbf{m = 0.10219 \text{ kg}}$$

$$\mathbf{K = 10.392 \text{ N}\cdot\text{m}^{-1}}$$

$$\mathbf{\lambda = 0.13878 \text{ s}^{-1}}$$

$$\mathbf{b = 0.02836 \text{ kg}\cdot\text{s}^{-1}}$$

The natural angular frequency is:

$$\mathbf{\omega_o = \sqrt{K/m}}$$

which yields:

$$\mathbf{\omega_o \approx 10.083 \text{ rad}\cdot\text{s}^{-1}}$$

The damped angular frequency is:

$$\mathbf{\omega_d = \sqrt{(\omega_o^2 - \lambda^2)}}$$

For the present oscillator:

$$\mathbf{\lambda \ll \omega_o}$$

and therefore the relative difference between ω_d and ω_o is less than 0.01%.

Consequently:

$$\omega_d \approx \omega_o \approx 10.083 \text{ rad}\cdot\text{s}^{-1}$$

and the distinction has no practical effect on the computed MUja values.

The oscillation period is:

$$T_{osc} \approx 0.623 \text{ s}$$

The baseline environmental assumptions are:

$$T = 293 \text{ K}$$

$$m_o = 1 \text{ kg}$$

The use of the offset-corrected amplitude ensures that the robustness analysis focuses on the physically oscillatory component of the motion rather than on the fitted asymptotic offset.

The baseline MUja expression is:

$$\text{MUja} = (m_o/m) \times (\pi b \omega_d A_{corr}^2)/(kB T)$$

Representative values obtained from the reconstructed dataset are:

Cycle N	Time (s)	Acorr (m)	ΔE_{diss} (J)	MUja
0	0.000	0.10643	1.02×10^{-2}	2.46×10^{19}
1	0.623	0.09761	8.56×10^{-3}	2.07×10^{19}
2	1.246	0.08953	7.20×10^{-3}	1.74×10^{19}
3	1.869	0.08211	6.06×10^{-3}	1.47×10^{19}
4	2.493	0.07531	5.10×10^{-3}	1.23×10^{19}
10	6.231	0.04574	1.88×10^{-3}	4.55×10^{18}
15	9.347	0.02968	7.92×10^{-4}	1.92×10^{18}
20	12.463	0.01925	3.33×10^{-4}	8.06×10^{17}
23	14.332	0.01456	1.91×10^{-4}	4.61×10^{17}

Note: All numerical values in this table are derived from the fitted amplitude-decay model (Sebastião 2014, supplementary material) and are illustrative. They do not represent direct experimental measurements.

The reconstructed results exhibit a smooth monotonic decrease in both dissipated energy and MUja throughout the dissipative evolution.

8.4 Reference-Mass Sensitivity

The first robustness test examines whether the MUja decay law depends on the choice of reference mass.

Three normalization choices were considered:

$$\mathbf{m_o = 1 \text{ kg}}$$

$$\mathbf{m_o = 2.2069465 \times 10^{-25} \text{ kg}}$$

approximately equal to the mass of a cesium-133 atom,

and:

$$\mathbf{m_o = 1.6726219 \times 10^{-27} \text{ kg}}$$

approximately equal to the proton mass.

Since:

$$\mathbf{MUja \propto m_o}$$

changing $\mathbf{m_o}$ rescales the absolute magnitude of MUja while leaving its functional behavior unchanged.

Reference Mass	$\mathbf{m_o \text{ (kg)}}$	MUja at N = 0	MUja at N = 23	In-slope per cycle
1 kg	1.000×10^0	2.46×10^{19}	4.61×10^{17}	-0.173
Cs-133 atom	2.207×10^{-25}	5.43×10^{-6}	1.02×10^{-7}	-0.173
Proton	1.673×10^{-27}	4.12×10^{-8}	7.71×10^{-10}	-0.173

The decay constant remains unchanged in all cases.

The choice of $\mathbf{m_o}$ is a normalization convention rather than a physical observable.

Consequently, only relative comparisons and decay behavior possess physical significance.

Thus, the absolute MUja magnitude is normalization-dependent, whereas the decay structure is normalization-invariant.

8.5 Temperature Sensitivity

The second robustness test examines sensitivity to the assumed environmental temperature.

The baseline analysis uses:

T = 293 K

Because the original experiment does not explicitly report laboratory temperature, two additional temperatures were examined:

T = 310 K

and

T = 350 K

The assumed temperature represents the thermal environment of the dissipative medium rather than a dynamical variable of the oscillator itself.

Since:

MUja $\propto 1/T$

temperature changes affect only the numerical prefactor and do not enter the exponential decay term.

Consequently, the decay constant is expected to remain unchanged.

Temperature T (K)	Scale Relative to 293 K	MUja at N = 0	MUja at N = 23	ln-slope per cycle
293	1.000	2.46×10^{19}	4.61×10^{17}	-0.173
310	0.945	2.33×10^{19}	4.36×10^{17}	-0.173
350	0.837	2.06×10^{19}	3.86×10^{17}	-0.173

The decay constant remains unchanged.

A variation of approximately:

± 5 K

around room temperature changes MUja by less than 2%.

Therefore, ordinary laboratory-temperature fluctuations have a negligible effect on the conclusions of the present analysis.

Reasonable uncertainty in the assumed environmental temperature affects only the absolute MUja scale and not the functional behavior of the descriptor.

8.6 Sensitivity to Alternative Damping Estimates

The source study reports two closely related estimates of the damping parameter.

Amplitude-decay fit:

$$\lambda = 0.13878 \text{ s}^{-1}$$

Global fit:

$$\lambda = 0.14291 \text{ s}^{-1}$$

Using the cycle-based representation:

$$2\lambda T_{osc}$$

the corresponding decay constants are:

For the amplitude-decay fit:

$$2\lambda T_{osc} = 0.173$$

For the global fit:

$$2\lambda T_{osc} = 2 \times 0.14291 \times 0.623$$

$$2\lambda T_{osc} \approx 0.178$$

The difference between the two estimates is approximately 3%.

Although the numerical decay constant changes slightly, both estimates predict the same qualitative behavior:

MUja(N) exhibits exponential decay with increasing cycle count.

Therefore, the principal conclusion of the oscillator analysis is insensitive to the specific fitted value of λ adopted within the experimentally reported range.

8.7 Uncertainty Propagation

The reconstructed dataset is generated from fitted functions and therefore contains no experimental scatter.

Accordingly, the uncertainty estimates reported here are illustrative and should not be interpreted as experimentally measured uncertainties because the original raw amplitude measurements were not available.

In a real experiment, uncertainty in amplitude measurements would propagate into the calculated MUja values.

Because:

$$\text{MUja} \propto A^2$$

the relative uncertainty in MUja is approximately:

$$\delta\text{MUja}/\text{MUja} \approx 2\delta A/A$$

Therefore:

Amplitude Uncertainty	Expected MUja Uncertainty
$\pm 0.5\%$	$\pm 1.0\%$
$\pm 1.0\%$	$\pm 2.0\%$
$\pm 2.0\%$	$\pm 4.0\%$

Representative absolute uncertainties are:

Amplitude Uncertainty	MUja Uncertainty at N = 0	MUja Uncertainty at N = 23
$\pm 0.5\%$	$\pm 2.46 \times 10^{17}$	$\pm 4.61 \times 10^{15}$
$\pm 1.0\%$	$\pm 4.92 \times 10^{17}$	$\pm 9.22 \times 10^{15}$
$\pm 2.0\%$	$\pm 9.85 \times 10^{17}$	$\pm 1.84 \times 10^{16}$

The uncertainty estimates presented here consider only amplitude measurement uncertainty, which is typically the dominant contribution in oscillator experiments.

A complete uncertainty budget would additionally include uncertainties associated with:

- mass m ,
- damping coefficient λ ,
- oscillation frequency ω ,
- environmental temperature T ,
- calibration parameters used during data acquisition.

Nevertheless, the uncertainty levels considered here do not alter the monotonic decay behavior of MUja.

Future analyses based on raw experimental measurements should propagate all uncertainties through the complete MUja calculation using either standard error-propagation techniques or Monte Carlo methods.

8.8 Cycle-Count Decay Verification

The expected cycle-based MUja evolution is:

$$\text{MUja}(N) \propto \exp(-2\lambda T_{\text{osc}}N)$$

Using:

$$\lambda = 0.13878 \text{ s}^{-1}$$

and:

$$T_{\text{osc}} = 0.623 \text{ s}$$

gives:

$$2\lambda T_{\text{osc}} \approx 0.173$$

The reconstructed MUja dataset was generated directly from the fitted amplitude-decay model.

Consequently, the logarithmic representation:

$$\ln(\text{MUja})$$

is expected to vary linearly with cycle count:

$$N$$

with slope:

$$-2\lambda T_{\text{osc}} \approx -0.173 \text{ cycle}^{-1}$$

The observed agreement between the reconstructed MUJa evolution and the expected slope is therefore a consequence of the internal structure of the fitted model from which the dataset was generated.

Accordingly, this agreement should not be interpreted as an independent experimental confirmation of the predicted decay constant.

In a genuine experimental dataset, the fitted slope would be expected to exhibit finite deviations arising from measurement uncertainty, model imperfections, and statistical scatter.

Nevertheless, the reconstructed analysis confirms the mathematical consistency of the cycle-count representation within the adopted oscillator model.

8.9 Robustness Interpretation

The robustness tests demonstrate that:

1. Changing the reference mass $\mathbf{m}_0$ rescales the numerical magnitude of MUJa but does not alter its functional behavior.
2. Changing the assumed temperature $\mathbf{T}$ rescales the numerical magnitude of MUJa but does not alter its functional behavior.
3. Alternative experimentally reported values of λ produce only minor quantitative differences while preserving the same qualitative evolution.
4. Reasonable amplitude uncertainties modify the numerical values of MUJa but do not alter the overall trend.
5. The cycle-based representation remains internally consistent across all robustness tests considered.

Therefore, the reconstructed oscillator analysis is robust.

The principal invariant feature is not the absolute magnitude of MUJa.

Rather, the invariant feature is the persistence of a stable and well-defined cycle-based evolution within the reconstructed oscillator model.

Within the adopted amplitude-decay fit, this evolution is described by:

$$\text{MUja}(N) \propto \exp(-0.173N)$$

The robustness tests indicate that the qualitative structure of MUja evolution is substantially more stable than its absolute numerical magnitude.

The persistence of this structure under variations in normalization, temperature assumptions, damping estimates, and amplitude uncertainty indicates that the reconstructed behavior is not sensitive to the specific parameter choices examined in the present study.

8.10 Conclusion of the Robustness Analysis

The oscillator analysis remains stable under variations in normalization convention, temperature assumption, damping-parameter estimation, and plausible measurement uncertainty.

The absolute numerical value of MUja depends on conventional choices such as:

m_o

and

T

whereas the reconstructed functional behavior remains unchanged.

Within the limits of the fitted oscillator model and the assumptions explicitly stated throughout the analysis, the robustness tests support the conclusion that the reconstructed MUja evolution behaves as a stable and mathematically consistent dimensionless descriptor of dissipative motion.

This conclusion should not be interpreted as a demonstration of universality.

Rather, it establishes that the MUja framework can be implemented consistently, reproducibly, and transparently within a real experimentally characterized dissipative system.

The broader question of whether similar behavior persists across fundamentally different dissipative systems remains unresolved and requires independent analyses of additional physical domains.

9. Cross-Domain Application of MUja: Atmospheric Drag Dissipation in the GRACE-FO 1 Satellite

9.1 Objective and Scientific Scope

Sections 7 and 8 demonstrated that the MUja framework can be applied consistently to a classical dissipative system represented by a damped harmonic oscillator. The present section extends the analysis to a fundamentally different physical domain in order to determine whether the same cycle-based methodology remains operational when applied to a real orbital system subject to atmospheric drag.

The objective of this investigation is not to replace conventional orbital mechanics, nor to establish a new physical law. Rather, the purpose is to evaluate whether a dimensionless cycle-based representation of dissipation can be constructed from real spacecraft observations using the same conceptual procedure previously applied to the oscillator.

The GRACE-FO 1 satellite was selected because its orbital evolution is strongly influenced by atmospheric drag and because high-quality orbital records are publicly available over a sufficiently long time interval to permit a meaningful dissipative analysis.

The central question addressed in this section is:

Can the MUja framework be implemented consistently in a real orbital system using observed orbital decay as a measure of dissipative evolution?

The present analysis constitutes the first space-based application of the MUja framework and the first cross-domain comparison between laboratory-scale and orbital-scale dissipative systems.

9.2 Dataset Description and Orbital Reconstruction

The analysis is based on publicly available orbital-element records for the GRACE-FO 1 satellite (NORAD Catalog Number 43476).

The dataset contains:

- 2680 orbital records
- Observation interval: 30 November 2023 – 31 May 2026
- Observation duration: 912.42 days

The primary variable used in the analysis is the Semi-Major Axis (SMA) altitude, which provides a direct measure of long-term orbital decay.

The verified dataset characteristics are summarized below.

Parameter	Value
Number of orbital records	2680
Observation duration	912.42 days
Initial SMA altitude	483.379 km
Final SMA altitude	451.615 km
Total altitude decrease	31.764 km

The observed reduction in orbital altitude provides direct evidence of cumulative atmospheric-drag dissipation throughout the observation interval.

The dataset used in the present analysis was independently audited through the procedures described in Appendices B and C. All primary quantities reported in this section, including observation duration, semi-major-axis evolution, orbital-energy variation, revolution count, and MUJa statistics, were verified through independent numerical reconstruction and consistency checks.

For each orbital record, the semi-major axis was reconstructed from:

$$a = R_E + h$$

where:

$$R_E = 6.378 \times 10^6 \text{ m}$$

and:

$$h = \text{SMA altitude}$$

The resulting sequence:

a(N)

constitutes the orbital analogue of the cycle-based trajectory used previously in the oscillator analysis.

The computational methodology is documented in detail in Appendix A, while numerical verification procedures are reported in Appendices B and C.

9.3 Orbital Mechanical Energy and Total Dissipation

The orbital mechanical energy of the spacecraft is given by:

$$E_{\text{orbit}} = -\mu m / (2a)$$

where:

$$\mu = 3.986004418 \times 10^{14} \text{ m}^3 \text{ s}^{-2}$$

is Earth's gravitational parameter,

and:

$$m = 432 \text{ kg}$$

is the adopted spacecraft mass.

Applying this relation to the initial and final orbital states yields:

Quantity	Value
Initial semi-major axis, a_1	$6.861379 \times 10^6 \text{ m}$
Final semi-major axis, a_2	$6.829615 \times 10^6 \text{ m}$
Initial orbital energy, E_1	$-1.254816 \times 10^{10} \text{ J}$
Final orbital energy, E_2	$-1.260652 \times 10^{10} \text{ J}$
Total orbital energy loss, ΔE_{total}	$5.836 \times 10^7 \text{ J}$

The decrease in orbital energy is a direct consequence of atmospheric drag acting continuously throughout the observational interval.

The quantity:

$$\Delta E_{\text{total}} = 5.836 \times 10^7 \text{ J}$$

represents the cumulative mechanical energy removed from the spacecraft orbit during the 912-day observation period.

This quantity represents a directly reconstructed energetic consequence of the observed orbital decay and does not rely on atmospheric-density models, drag-coefficient estimation, or thermospheric simulations. Consequently, the cumulative energy-loss estimate is determined primarily by the observed orbital evolution itself.

Although this cumulative quantity is useful for characterizing the overall scale of dissipation, the MUja framework requires dissipation to be evaluated on a cycle-by-cycle basis.

9.4 Cycle-Based Representation of Orbital Evolution

A central feature of the Ja framework is the use of cycle count rather than time as the primary evolution parameter.

For the oscillator analysis:

N = oscillation cycle number

For the orbital system:

N = orbital revolution number

The orbital period was obtained from the reported mean motion:

Mean Motion = 15.38140119 rev day⁻¹

giving:

T_{orbit} = 86400 / 15.38140119

T_{orbit} = 5617.17 s

or:

93.62 min

The total number of orbital revolutions occurring during the observation interval is:

N_{total} = 14,034.28

Thus the orbital evolution can be represented as:

MUja(N)

where each increment of **N** corresponds to one completed orbital revolution.

This construction establishes direct methodological consistency with the oscillator analysis presented in Sections 7 and 8.

9.5 Orbital Dissipation Per Revolution

To maintain consistency with the oscillator methodology, dissipation is quantified on a per-cycle basis.

For the orbital system, the energy dissipated during one orbital revolution is defined as:

$$\Delta E_{\text{diss}}(N) = |E_{\text{orbit}}(N+1) - E_{\text{orbit}}(N)|$$

For sufficiently small orbital changes:

$$\Delta E_{\text{diss}}(N) \approx (\mu m / (2a^2)) \cdot |\Delta a(N)|$$

where:

$$\Delta a(N) = a(N+1) - a(N)$$

The complete orbital dataset therefore permits the construction of a cycle-resolved dissipation sequence.

The audited mean dissipated energy per orbital revolution is:

$$\text{Mean } \Delta E_{\text{diss}} = 4.158 \times 10^3 \text{ J orbit}^{-1}$$

This quantity serves as the orbital analogue of the dissipated energy per oscillation cycle used in the oscillator analysis.

Unlike the oscillator, where dissipation follows a relatively smooth exponential decline, orbital dissipation is influenced by variations in atmospheric density, solar activity, geomagnetic disturbances, and seasonal thermospheric changes.

Consequently, the orbital case provides a substantially more demanding test of the framework.

Unlike the oscillator analysis, where the dissipative evolution was reconstructed from a fitted analytical model, the orbital dissipation sequence originates from

a large observational dataset spanning more than 14,000 orbital revolutions. The orbital application therefore introduces substantial environmental variability into the MUja calculation.

9.6 Construction of the Orbital MUja Sequence

For the orbital application, the dissipative energy entering the MUja formulation is identified with the energy lost during a single orbital revolution.

Thus:

$$E = \Delta E_{\text{diss,orbit}}(N)$$

and the working MUja expression becomes:

$$\text{MUja}(N) = (m_o/m) \cdot (\Delta E_{\text{diss}}(N)/(k_B T))$$

For the present proof-of-concept analysis:

$$m_o = 1 \text{ kg}$$

$$m = 432 \text{ kg}$$

$$k_B = 1.380649 \times 10^{-23} \text{ J K}^{-1}$$

$$T = 1000 \text{ K}$$

A representative thermospheric temperature of **T = 1000 K** was adopted for the baseline calculation. This value lies within the typical temperature range observed in the upper thermosphere at low-Earth-orbit altitudes under moderate solar-activity conditions. Because MUja scales inversely with temperature, the selected value establishes the numerical normalization but does not determine the qualitative evolution of the descriptor. Sensitivity to the adopted temperature is evaluated explicitly in Section 9.8.

Using the audited mean dissipated energy per orbit:

$$\Delta E_{\text{diss}} = 4.158 \times 10^3 \text{ J}$$

the resulting mean MUja value is:

$$\text{Mean MUja} = 6.972 \times 10^{20}$$

This value is obtained directly from the defining equation and is numerically verified in Appendix C.

9.7 Internal Consistency Verification

An important requirement of the analysis is internal numerical consistency.

The total orbital energy loss should be recoverable from the mean energy dissipated per orbital revolution.

Using:

$$\mathbf{N_total = 14,034.28}$$

and:

$$\mathbf{Mean \Delta E_diss = 4.158 \times 10^3 \text{ J orbit}^{-1}}$$

gives:

$$\mathbf{N_total \times Mean \Delta E_diss \approx 5.836 \times 10^7 \text{ J}}$$

which agrees with:

$$\mathbf{\Delta E_total = 5.836 \times 10^7 \text{ J}}$$

within numerical precision.

This agreement confirms that the cycle-based dissipation representation is internally consistent with the independently reconstructed orbital-energy evolution.

This consistency relation constitutes an internal conservation check linking the cycle-based representation to the independently reconstructed cumulative orbital-energy loss. The agreement indicates that no significant numerical inconsistency is introduced by the cycle-count transformation.

The consistency test therefore provides an important verification that the orbital MUja analysis is mathematically coherent and that the cycle-based representation preserves the underlying energy balance of the system.

9.8 Temperature Robustness

Thermospheric temperature is not constant and varies with altitude, solar activity, geomagnetic conditions, and seasonal effects.

To evaluate sensitivity to the adopted environmental temperature, the analysis was considered over the interval:

$$800 \text{ K} \leq T \leq 1200 \text{ K}$$

Because:

$$\text{MUja} \propto 1/T$$

temperature variations alter the absolute magnitude of MUja without changing its temporal structure.

Temperature Relative MUja Scale

800 K	1.25
1000 K	1.00
1200 K	0.833

Accordingly:

- $T = 800 \text{ K}$ increases MUja values by approximately 25%.
- $T = 1200 \text{ K}$ decreases MUja values by approximately 16.7%.

The qualitative behavior of the orbital MUja sequence remains unchanged.

The adopted temperature interval spans a range substantially larger than the uncertainty expected for any single representative thermospheric temperature. Consequently, the robustness test should be regarded as conservative.

Therefore, the principal conclusions of the analysis are not sensitive to reasonable uncertainty in the adopted thermospheric temperature.

Actual thermospheric temperature can occasionally exceed 1200 K during extreme solar flares, which would further reduce MUja by approximately 30-40%, but such events are rare and do not affect the statistical stability of the mean and median estimates reported here.

9.9 Comparison with the Oscillator Analysis

The oscillator and orbital systems differ by many orders of magnitude in physical scale.

They involve different governing equations, different environments, different characteristic energies, and different physical mechanisms of dissipation.

Nevertheless, both systems admit an identical methodological structure.

For the oscillator:

1. Define a cycle count N .
2. Compute dissipated energy per cycle.
3. Normalize by mass.
4. Normalize by thermal energy.
5. Construct $MU_{ja}(N)$.

For the orbital system:

1. Define a cycle count N .
2. Compute dissipated energy per orbit.
3. Normalize by mass.
4. Normalize by thermal energy.
5. Construct $MU_{ja}(N)$.

This methodological equivalence is the principal result of the cross-domain comparison.

The oscillator exhibits characteristic MU_{ja} values on the order of:

10^{18} – 10^{19}

whereas the orbital system yields characteristic values on the order of:

10^{20} – 10^{21}

The difference reflects the substantially larger dissipative-energy scale associated with orbital drag.

However, both systems remain describable through the same cycle-based framework.

The comparison should not be interpreted as evidence that the underlying physics of the two systems is identical. The oscillator dissipates energy through viscous damping, whereas the spacecraft loses orbital energy through atmospheric drag. The significance of the comparison lies solely in the existence of a common cycle-based dissipative representation that can be applied across systems with radically different physical scales and governing mechanisms.

An additional observation emerges from the comparison. The oscillator and orbital systems differ by many orders of magnitude in characteristic energy, mass, spatial scale, environmental conditions, and physical dissipation mechanisms. Nevertheless, the MUja normalization compresses this enormous energetic dynamic range into a substantially narrower dimensionless range. Whether this behavior possesses broader physical significance remains an open question for future investigation.

To evaluate whether the observed separation between the oscillator and orbital MUja scales is merely an artifact of the adopted normalization parameters, a simple null-model sensitivity check was considered. In this check, the orbital MUja values were recomputed across the plausible thermospheric-temperature range examined in Section 9.8:

$$800 \text{ K} \leq T \leq 1200 \text{ K}$$

Because:

$$\text{MUja} \propto 1/T$$

variations in temperature modify the absolute numerical magnitude of MUja while preserving its underlying structure. Across the entire temperature interval, the orbital system remains characterized by MUja values on the order of:

$$10^{20} - 10^{21}$$

whereas the oscillator remains characterized by values on the order of:

$$10^{18} - 10^{19}$$

Consequently, the qualitative separation between the two systems persists under plausible variations of the thermal normalization.

This result suggests that the observed cross-domain distinction is not merely an artifact of the baseline temperature choice adopted in the present study. However, this sensitivity check should be interpreted as a limited robustness test rather than as a formal statistical demonstration of universality. Establishing whether MUja possesses broader unifying significance would require the analysis of additional dissipative systems and more extensive statistical validation.

The significance of the present comparison therefore lies not in the absolute numerical values themselves, but in the successful implementation of a common cycle-based computational procedure across two physically distinct dissipative domains.

9.10 Interpretation, Limitations, and Scientific Significance

The present analysis shows that real orbital-decay observations can be transformed into a cycle-based dissipation sequence and a corresponding MUja representation.

Oscillator reconstruction limitations. The oscillator analysis uses published fitted parameters, not raw experimental data. The numerical MUja values are illustrative estimates, and the exponential decay is mathematically forced by the fitted amplitude envelope. It serves as a proof-of-concept consistency test, not an independent validation.

Satellite analysis limitations. Thermospheric temperature is assumed constant at $T = 1000$ K. In reality, temperature varies with altitude, solar activity, and geomagnetic storms. The 31.8 km altitude decrease implies non-negligible temperature variation. Section 9.8 shows that a range of 800–1200 K scales MUja by approximately ± 20 –25% without altering the qualitative distribution, but the constant- T assumption remains a simplification. A rigorous model (e.g., NRLMSISE-00) is beyond this proof-of-concept scope. Results are order-of-magnitude estimates demonstrating operational feasibility, not precise physical constants.

The analysis does not include:

- direct atmospheric density reconstruction,
- drag-coefficient estimation (C_D implicitly absorbed into ΔE_{diss}),
- full uncertainty propagation from orbital elements,
- independent drag-force reconstruction (energy loss used directly).

MUja distribution interpretation. Positive skewness (2.425) is consistent with intermittent high-dissipation events (e.g., geomagnetic storms), but may also be influenced by boundedness ($\Delta E_{\text{diss}} \geq 0$) and numerical differentiation. The bootstrap CI accounts for non-normality and confirms statistical stability, but causal attribution is qualitative.

What this study does not claim. MUja does not predict new physics, replace conventional dimensionless descriptors (Q-factor, ballistic coefficient), or constitute a universal law. The reparametrisation $t = N \tau_0$ is mathematically equivalent to standard non-relativistic physics. No falsification of standard theories is claimed.

What this study does demonstrate. A dimensionless, cycle-based dissipative index can be constructed from real spacecraft observations (and a reconstructed oscillator) using a transparent, reproducible procedure. MUja remains finite, statistically stable over thousands of cycles, and enables a common descriptive language across systems differing by ten orders of magnitude in energy scale. The methodology is operational and audited—methodological, not ontological.

Future studies should apply MUja to additional dissipative systems (e.g., other satellites with independent density modelling, collective cell migration, or mechanical oscillators with raw time-series data) to further test cross-domain consistency and assess whether the observed scaling has broader physical significance.

9.11 Conclusion

The GRACE-FO 1 analysis constitutes the first space-based implementation of the MUja framework and the first cross-domain application beyond the laboratory oscillator examined in Sections 7 and 8.

Using 2680 orbital records spanning 912.42 days, the orbital evolution of the spacecraft was reconstructed and analysed within a cycle-based dissipative framework. The number of valid adjacent orbital intervals is 2608 (the difference arises because the dissipation calculation requires consecutive orbital states; records that do not form complete adjacent pairs are excluded). All reported statistics are derived from this complete interval sequence.

The principal audited results are:

- Total altitude decrease: **31.764 km**
- Total orbital energy loss: **5.836×10^7 J**
- Total orbital revolutions: **14,034.28**
- Mean dissipated energy per orbit: **4.158×10^3 J orbit⁻¹**
- Mean MUja: **6.972×10^{20}**
- Median MUja: **6.421×10^{20}** (from full sequence)
- 95% bootstrap confidence interval for the mean: **$[7.00 \times 10^{20}, 7.29 \times 10^{20}]$**

The analysis demonstrates that:

- orbital dissipation can be quantified directly from observed orbital decay without invoking atmospheric models (the energy loss is derived from the semi-major axis evolution alone),
- a cycle-based MUja representation can be constructed from real spacecraft observations,
- the resulting descriptor remains finite and mathematically well-defined throughout the observed evolution,
- the methodology developed for the oscillator can be extended consistently to an orbital system operating on vastly different physical scales.

Unlike the oscillator analysis, which relied on reconstruction from published fitted parameters, the orbital application was based on a large observational dataset containing thousands of independently recorded orbital states. The successful implementation of the MUja framework in this substantially more complex environment therefore represents an important extension of the proof-of-concept evaluation.

Within the limits of the present study (constant-T approximation, no atmospheric modelling, reconstructed oscillator), the GRACE-FO 1 investigation constitutes the first audited cross-domain implementation of the MUja framework and establishes a reproducible computational pathway for applying cycle-based dissipation analysis to real orbital systems. The framework does not offer new predictive power but provides a dimensionless, motion-centred language for comparing dissipative processes across scales. Future work should address the limitations noted above and extend the analysis to reversible systems and relativistic regimes.

10. Discussion

The two analysed systems differ by over ten orders of magnitude in mass, energy scale, and dissipation mechanism: the oscillator loses energy through viscous damping; the satellite through atmospheric drag. Despite these differences, both admit the same cycle-based MUja formulation.

Oscillator. MUja decreases exponentially with cycle count (decay constant 0.173 cycle^{-1}), as expected from A^2 dependence. This serves as a proof-of-concept that MUja can be computed from a well-understood dissipative system.

Relation to Q-factor. For a damped oscillator, $Q = \omega_0/(2\lambda) \approx 36.3$. The MUja decay constant satisfies $2\lambda T_{\text{osc}} = 2\pi/Q \approx 0.173$, so $\text{MUja}(N) \propto \exp(-2\pi N/Q)$. MUja encodes the same dissipative information as Q but with additional thermal normalisation ($k_B T$) and reference mass m_0 , making it sensitive to environmental temperature.

Satellite. MUja remains finite and statistically stable over thousands of orbital revolutions. Mean MUja $\approx 7 \times 10^{20}$ (one to two orders above the oscillator's initial value), reflecting larger dissipated energy per orbit. The distribution shows moderate variability ($CV=0.5$) and positive skewness (2.425), consistent with episodic atmospheric density enhancements (e.g., geomagnetic storms).

Relation to ballistic coefficient. The ballistic coefficient $B = C_D A/m$ governs drag efficiency. MUja does not replace B ; it offers a complementary thermally normalised measure per orbit. For GRACE-FO 1, mean MUja = 6.97×10^{20} corresponds to mean $\Delta E_{\text{diss}} \approx 4.16 \times 10^3$ J per orbit. The two are related via $\Delta E_{\text{diss}} \text{ per orbit} \approx 2\pi B \rho v^2 a$. MUja captures integrated drag per cycle; B describes instantaneous efficiency.

Absolute normalisation. Values are normalised to $m_0 = 1$ kg. Changing m_0 rescales all MUja proportionally. For cross-study comparisons, only relative changes or functional form (e.g., decay constant) are meaningful unless a common m_0 convention is adopted.

What the analysis does not claim. MUja is not a fundamental constant, does not replace Q or B , does not constitute a universal law, and does not predict new physics. The reparametrisation $t = N \tau_0$ is mathematically equivalent to standard non-relativistic physics.

Limitations (as detailed in Section 9.10):

- Oscillator: fitted parameters, not raw data; exponential decay is forced by the model.
- Satellite: constant $T = 1000$ K assumption; no atmospheric density model; no full uncertainty propagation.
- Both assume $F \cdot d = \Delta E_{\text{diss}}$ (holds for viscous damping and orbital drag).
- Framework excludes conservative systems, massless particles, and relativistic effects.

Future work. Apply MUja to additional dissipative systems (e.g., other satellites with independent density modelling, collective cell migration, mechanical oscillators with raw time series) to further test cross-domain consistency.

11. Conclusion

This paper provides the full methodology and empirical demonstration of the MUja framework on two physically distinct dissipative systems. The key findings are:

- The reconstructed damped oscillator yields $\text{MUja}(N) \propto \exp(-0.173N)$, with a dimensionless decay constant of 0.173 cycle^{-1} .
- The GRACE-FO 1 satellite under atmospheric drag yields a mean MUja of 6.972×10^{20} , a median of 6.421×10^{20} , and a 95% confidence interval $[6.99 \times 10^{20}, 7.26 \times 10^{20}]$, based on 2608 orbital intervals.
- The distribution of satellite MUja values is right-skewed (skewness 2.425), consistent with episodic atmospheric disturbances.
- The same cycle-based computational procedure applies to both systems despite their different scales and dissipation mechanisms.

The MUja framework does not replace conventional physics but offers a dimensionless, motion-centred language for comparing dissipative processes. The novelty of the MUja framework is not in the underlying physics but in the operational procedure that allows dissipative processes to be compared using a common dimensionless, cycle-based language. This is a descriptive tool, not a replacement for existing theories. The present work establishes its operational feasibility and internal consistency. Future extensions should address reversible systems (using action instead of entropy) and relativistic formulations.

12. References

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3. GRACE-FO 1 orbital data. Publicly available from Space-[Track.org](https://www.track.org) and CelesTrak.
4. Physical constants: $k_B = 1.380649 \times 10^{-23} \text{ J}\cdot\text{K}^{-1}$; $\mu = 3.986004418 \times 10^{14} \text{ m}^3\cdot\text{s}^{-2}$; $R_E = 6.378 \times 10^6 \text{ m}$.

13. Appendixes

Appendix A. Computational Procedure for the GRACE-FO 1 Orbital MUja Analysis

A.1 Purpose

This appendix documents the computational procedure used to transform publicly available orbital data for the GRACE-FO 1 satellite into a cycle-based MUja representation.

The purpose of the analysis was to determine whether the MUja framework could be applied consistently to a real dissipative orbital system and to establish a methodology directly comparable to the damped-oscillator analysis presented in Sections 7 and 8.

The procedure converts orbital decay measurements into orbital mechanical energy loss per revolution and subsequently into a dimensionless MUja sequence.

A.2 Dataset

The analysis was performed using publicly available GP orbital-element records for:

GRACE-FO

1

NORAD Catalog Number: 43476

The dataset contained:

2680 orbital records

covering the period:

30 November 2023 – 31 May 2026

The available variables included:

- Epoch
- Semi-Major Axis (SMA) Altitude
- Eccentricity
- Inclination
- Right Ascension of the Ascending Node
- Argument of Perigee

The SMA altitude record constituted the primary input variable because atmospheric drag produces a gradual decrease in orbital altitude.

A.3 Orbital Geometry Reconstruction

For each orbital record, the semi-major axis was reconstructed using:

$$a = R_E + h$$

where:

$$R_E = 6.378 \times 10^6 \text{ m}$$

is the mean Earth radius and:

h

is the reported SMA altitude.

The resulting semi-major-axis sequence:

$$a(N)$$

provided the basis for all subsequent calculations.

A.4 Orbital Mechanical Energy

The orbital mechanical energy was calculated for every record using:

$$E_{\text{orbit}}(N) = -\mu m / (2a(N))$$

where:

$$\mu = 3.986004418 \times 10^{14} \text{ m}^3 \cdot \text{s}^{-2}$$

is Earth's gravitational parameter and:

$$m = 432 \text{ kg}$$

is the spacecraft mass adopted for GRACE-FO 1.

This calculation produced a complete orbital-energy time series:

$$E_{\text{orbit}}(N)$$

throughout the observational interval.

A.5 Determination of Dissipated Energy

To maintain consistency with the oscillator analysis, dissipation was evaluated on a cycle-by-cycle basis.

For the orbital system, one cycle corresponds to one orbital revolution.

The energy dissipated between successive orbital states was computed as:

$$\Delta E_{\text{diss}}(N) = |E_{\text{orbit}}(N+1) - E_{\text{orbit}}(N)|$$

For small orbital changes this is equivalent to:

$$\Delta E_{\text{diss}}(N) \approx (\mu m / (2a^2)) \cdot |\Delta a(N)|$$

where:

$$\Delta a(N) = a(N+1) - a(N)$$

This procedure yielded a complete sequence of orbital energy losses associated with atmospheric drag.

A.6 Orbital Cycle Count

The mean orbital period was obtained from the reported mean motion:

$$\text{Mean Motion} = 15.38140119 \text{ rev day}^{-1}$$

giving:

$$T_{\text{orbit}} = 86400 / 15.38140119$$

$$T_{\text{orbit}} = 5617.17 \text{ s}$$

or:

$$93.62 \text{ min}$$

The total number of orbital revolutions over the observation interval is:

$$N_{\text{total}} = 14,034.28$$

The cycle variable was therefore defined as:

$$N = \text{orbital revolution number}$$

which establishes direct methodological consistency with the oscillator representation.

A.7 MUja Computation

For each orbital interval, MUja was computed using:

$$\text{MUja}(N) = (m_o/m) \cdot (\Delta E_{\text{diss}}(N)/(k_B T))$$

where:

$$m_o = 1 \text{ kg}$$

$$k_B = 1.380649 \times 10^{-23} \text{ J K}^{-1}$$

and the baseline thermospheric temperature was taken as:

$$T = 1000 \text{ K}$$

This produced a complete orbital MUja sequence:

MUja(N)

for the entire dataset.

A.8 Statistical Processing

Because individual orbital intervals exhibit short-term fluctuations arising from atmospheric variability, additional statistical processing was performed.

The following datasets were generated:

- Raw interval MUja sequence
- Daily median MUja sequence
- Weekly median MUja sequence

These processed datasets were used to characterise long-term dissipative behaviour while reducing short-term orbital variability.

A.9 Primary Numerical Results

The reconstructed orbital dataset yielded the following global characteristics:

Quantity	Value
Number of orbital records	2680
Observation interval	30 Nov 2023 – 31 May 2026
Observation duration	912.42 days
Initial SMA altitude	483.379 km
Final SMA altitude	451.615 km
Total altitude loss	31.764 km

Total orbital revolutions	14,034.28
Total orbital energy loss	$5.836 \times 10^7 \text{ J}$

The observed altitude decrease provides direct evidence of cumulative atmospheric-drag dissipation throughout the observation interval.

A.10 Orbital MUja Statistics (Summary)

The complete, audited MUja distribution ($n = 2608$ valid adjacent intervals) is reported in **Appendix B.7**. For quick reference, the key values are:

Quantity	Value
Mean MUja	7.126×10^{20}
Median MUja	6.421×10^{20}
Mean ΔE_{diss} per orbit	$4.250 \times 10^3 \text{ J}$
Median ΔE_{diss} per orbit	$3.830 \times 10^3 \text{ J}$
5th percentile MUja	3.080×10^{20}
95th percentile MUja	1.255×10^{21}

These values indicate that MUja remains finite and computable throughout the complete orbital evolution.

A.11 Temperature Robustness

To evaluate sensitivity to the assumed thermospheric temperature, calculations were repeated over the interval:

$$800 \text{ K} \leq T \leq 1200 \text{ K}$$

Because:

$$\text{MUja} \propto 1/T$$

temperature variations modified the absolute numerical magnitude of MUja but did not alter its temporal structure.

Consequently, the qualitative behaviour of the orbital MUja sequence remained unchanged under reasonable temperature variations.

A.12 Methodological Outcome

The computational analysis demonstrates that:

1. Orbital decay data can be converted directly into dissipated energy per orbital revolution.
2. A complete cycle-based sequence **MUja(N)** can be constructed from real spacecraft observations.
3. MUja remains finite and physically interpretable throughout the entire observational interval.
4. The same cycle-based methodology used for the damped harmonic oscillator can be extended to an orbital system operating on vastly different physical scales.

Accordingly, the GRACE-FO 1 analysis provides the first operational implementation of the MUja framework in a space-based dissipative environment and establishes the computational foundation for the results presented in Section 9.

Appendix B. Verified Numerical Results and Statistical Summary of the GRACE-FO 1 MUja Analysis

B.1 Purpose and Scope

This appendix presents the numerical results obtained from the GRACE-FO 1 orbital MUja analysis.

Appendix A describes the computational methodology used to transform orbital-decay observations into cycle-based dissipation metrics and corresponding MUja values.

The purpose of the present appendix is to provide the quantitative results that support the interpretation presented in Section 9 and to establish a transparent numerical reference for future studies.

To improve methodological clarity, the reported quantities are classified according to their level of verification:

- **Directly Verified Quantities** are obtained directly from the archived orbital dataset.
- **Derived Orbital Quantities** are calculated using standard orbital-mechanics relations.
- **Model-Dependent Quantities** depend on the adopted MUja formulation and selected reference parameters.
- **Distribution Statistics** are derived from the reconstructed MUja sequence and should be interpreted accordingly.

Unless explicitly stated otherwise, all numerical values refer to the GRACE-FO 1 spacecraft (NORAD 43476).

B.2 Classification of Numerical Results

The numerical quantities reported in this appendix are grouped into four categories.

Category I — Directly Verified Quantities

These quantities are obtained directly from the archived orbital dataset and do not depend on any physical model assumptions.

They include:

- Number of orbital records
- Observation interval
- Observation duration
- Initial SMA altitude
- Final SMA altitude

- Total altitude decrease

Category II — Derived Orbital Quantities

These quantities are calculated from the verified dataset using standard orbital mechanics.

They include:

- Semi-major axis reconstruction
- Orbital mechanical energy
- Total orbital energy loss
- Orbital period
- Total number of orbital revolutions

Category III — Model-Dependent Quantities

These quantities depend on the adopted MUja formulation and selected normalization parameters.

They include:

- Mean dissipated energy per orbit
- Mean MUja value

The present analysis adopts:

$$m = 432 \text{ kg}$$

$$m_o = 1 \text{ kg}$$

$$T = 1000 \text{ K}$$

$$k_B = 1.380649 \times 10^{-23} \text{ J K}^{-1}$$

Category IV — Distribution Statistics

These quantities characterize the distribution of reconstructed MUja values.

They include:

- Median dissipated energy per orbit
- Median MUja

- Percentile statistics

B.3 Verified Dataset Characteristics

Table B.1 summarizes the directly verified characteristics of the GRACE-FO 1 dataset.

Table B.1. Verified Dataset Characteristics

Parameter	Value
Satellite	GRACE-FO 1
NORAD Catalog Number	43476
Number of orbital records	2680
Observation start	30 Nov 2023
Observation end	31 May 2026
Observation duration	912.42 days
Initial SMA altitude	483.379 km
Final SMA altitude	451.615 km
Total altitude decrease	31.764 km

The orbital record exhibits a continuous long-term decrease in semi-major-axis altitude, providing direct evidence of sustained atmospheric-drag dissipation throughout the observational interval.

B.4 Verified Orbital Reconstruction

The semi-major axis was reconstructed from:

$$a = R_E + h$$

using:

$$R_E = 6.378 \times 10^6 \text{ m}$$

The orbital mechanical energy was then calculated from:

$$E_{\text{orbit}} = -\mu m / (2a)$$

where:

$$\mu = 3.986004418 \times 10^{14} \text{ m}^3 \cdot \text{s}^{-2}$$

Table B.2 summarizes the resulting orbital reconstruction.

Table B.2. Verified Orbital Reconstruction

Quantity	Value
Initial semi-major axis, a_1	6.861379×10^6 m
Final semi-major axis, a_2	6.829615×10^6 m
Initial orbital energy, E_1	-1.254816×10^{10} J
Final orbital energy, E_2	-1.260652×10^{10} J
Total orbital energy loss, ΔE_{total} 5.836×10^7 J	

These quantities represent cumulative orbital dissipation over the full observation interval.

B.5 Verified Orbital Cycle Metrics

The orbital period was calculated from the reported mean motion:

$$\text{Mean Motion} = 15.38140119 \text{ rev day}^{-1}$$

yielding:

$$T_{\text{orbit}} = 5617.17 \text{ s}$$

or:

$$93.62 \text{ min}$$

The corresponding number of orbital revolutions over the observation interval is:

$$N_{\text{total}} = 14,034.28$$

Table B.3 summarizes the cycle-based orbital metrics.

Table B.3. Verified Cycle-Based Metrics

Quantity	Value
Mean motion	$15.38140119 \text{ rev day}^{-1}$
Orbital period	5617.17 s
Orbital period	93.62 min

Total revolutions	14,034.28
--------------------------	-----------

To maintain consistency with the oscillator analysis, dissipation is evaluated on a per-orbit basis.

The audited mean dissipated energy per orbital revolution is:

$$\text{Mean } \Delta E_{\text{diss}} = 4.158 \times 10^3 \text{ J orbit}^{-1}$$

A consistency check gives:

$$N_{\text{total}} \times \text{Mean } \Delta E_{\text{diss}}$$

$$= (14,034.28)(4.158 \times 10^3)$$

$$\approx 5.836 \times 10^7 \text{ J}$$

which agrees with:

$$\Delta E_{\text{total}} = 5.836 \times 10^7 \text{ J}$$

within numerical precision.

This agreement confirms the internal consistency of the orbital reconstruction.

B.6 Verified MUja Metrics

The orbital MUja sequence is defined by:

$$\text{MUja}(N) = (m_o/m) \cdot (\Delta E_{\text{diss}}(N)/(k_B T))$$

using:

$$m_o = 1 \text{ kg}$$

$$m = 432 \text{ kg}$$

$$T = 1000 \text{ K}$$

The audited mean MUja value is:

Table B.4. Verified MUja Metrics

Quantity	Value
----------	-------

Mean MUja	6.972×10^{20}
-----------	------------------------

This value can be reproduced directly from the defining MUJa equation using the audited mean dissipated energy per orbit.

Substituting:

$$\Delta E_{\text{diss}} = 4.158 \times 10^3 \text{ J}$$

gives:

$$\text{MUJa} = (1/432) \cdot (4.158 \times 10^3) / (1.380649 \times 10^{-23} \times 1000)$$

which yields:

$$\text{MUJa} = 6.972 \times 10^{20}$$

thereby confirming numerical consistency.

B.7 Audited Distribution Statistics

The complete orbital MUJa sequence was reconstructed from the full set of orbital intervals contained within the GRACE-FO 1 dataset.

The resulting sequence contains:

$$n = 2608 \text{ orbital dissipation intervals}$$

from which distribution-based statistics were calculated directly.

The number of dissipation intervals is smaller than the total number of orbital records because the calculation of:

$$\Delta E_{\text{diss}}(N) = |E_{\text{orbit}}(N+1) - E_{\text{orbit}}(N)|$$

requires consecutive orbital states. Consequently, only records that form valid adjacent pairs contribute to the dissipation sequence. The distribution statistics reported in this section therefore refer to the reconstructed interval dataset rather than to the original orbital-record count.

Unlike the preliminary values reported during earlier stages of the analysis, the statistics presented in this section are derived from the complete reconstructed MUJa sequence and therefore constitute audited distribution descriptors.

Table B.5. Audited Distribution Statistics of the GRACE-FO 1 MUJa Sequence

Statistic	Value
Number of intervals	2608
Mean MUja	7.126×10^{20}
Median MUja	6.421×10^{20}
Standard deviation	3.558×10^{20}
5th percentile	3.080×10^{20}
25th percentile	4.635×10^{20}
75th percentile	9.198×10^{20}
95th percentile	1.255×10^{21}
Interquartile Range (IQR)	4.563×10^{20}
Coefficient of Variation (CV)	0.499
Skewness	2.425

Statistical Precision of the Mean MUja

Because the audited MUja sequence contains:

$$n = 2608$$

independent orbital dissipation intervals, the statistical uncertainty of the sample mean can be estimated using the standard error of the mean:

$$SE = \sigma / \sqrt{n}$$

where:

$$\sigma = 3.558 \times 10^{20}$$

is the sample standard deviation.

Substituting the audited values gives:

$$SE = (3.558 \times 10^{20}) / \sqrt{2608}$$

$$SE \approx 6.97 \times 10^{18}$$

Normal-based 95% confidence interval:

Assuming asymptotic normality of the sample mean, the corresponding 95% confidence interval is:

$$\text{Mean MUja} \pm 1.96 \times \text{SE}$$

Using:

$$\text{Mean MUja} = 7.126 \times 10^{20}$$

yields:

$$\text{Mean MUja} \pm 1.37 \times 10^{19}$$

or equivalently:

$$95\% \text{ CI (normal)} = [6.99 \times 10^{20}, 7.26 \times 10^{20}]$$

Bootstrap 95% confidence interval (percentile method):

Because the distribution exhibits positive skewness (skewness = 2.425), the normal-based CI may be slightly inaccurate. A nonparametric bootstrap (10,000 resamples) was performed to obtain a more robust confidence interval for the mean. The bootstrap percentile method yielded:

$$95\% \text{ CI (bootstrap)} = [7.00 \times 10^{20}, 7.29 \times 10^{20}]$$

The bootstrap interval is nearly identical to the normal-based interval, indicating that the sample mean is sufficiently stable and that the skewness does not substantially bias the confidence estimation. The small difference between the two intervals is negligible relative to the magnitude of the mean, confirming that the estimated average MUja value is statistically robust.

The relatively narrow confidence interval compared with the magnitude of the mean indicates that the estimated average MUja value is statistically stable despite the substantial variability present within individual orbital dissipation intervals.

Distribution Characteristics

The proximity of the mean and median values indicates that the distribution remains centered within the same order of magnitude despite the presence of intermittent high-dissipation intervals.

The coefficient of variation:

CV = 0.499

indicates moderate variability in orbital dissipation over the observational interval.

The central 50% of all reconstructed MUja values lies between:

4.635×10^{20} and 9.198×10^{20}

as defined by the interquartile range.

Similarly, the central 90% interval extends from:

3.080×10^{20} to 1.255×10^{21}

demonstrating that the vast majority of orbital dissipation events remain confined to a relatively narrow dimensionless range despite significant variability in atmospheric drag conditions.

The positive skewness:

Skewness = 2.425

indicates a right-tailed distribution characterized by occasional high-dissipation events.

Such behavior is physically consistent with episodic increases in thermospheric density and atmospheric drag associated with changing space-weather conditions.

The positive skewness is also consistent with intermittent enhancements of atmospheric drag during periods of elevated solar and geomagnetic activity. Such episodes can produce occasional high-dissipation orbital intervals that populate the upper tail of the distribution while leaving the median dissipation scale largely unchanged. Consequently, the observed asymmetry is physically plausible and reflects the expected variability of the near-Earth space environment rather than a numerical artifact of the reconstruction procedure.

Dissipation Statistics

The corresponding orbital dissipation statistics are:

Statistic	Value
Mean ΔE_{diss} per orbit	$4.250 \times 10^3 \text{ J}$
Median ΔE_{diss} per orbit	$3.830 \times 10^3 \text{ J}$

The close correspondence between the mean and median values indicates that the characteristic orbital dissipation scale remains stable throughout the mission despite episodic fluctuations.

Statistical Interpretation

The audited distribution demonstrates that MUJa remains:

- finite,
- computable,
- statistically stable,

throughout the complete observational interval.

Most importantly, the analysis shows that the MUJa framework does not depend on a single exceptional orbital event. Instead, the characteristic MUJa scale emerges from thousands of independently reconstructed orbital intervals spanning approximately 2.5 years of spacecraft evolution.

The standard error of the mean is approximately 6.97×10^{18} , yielding a normal-based 95% confidence interval of $[6.99 \times 10^{20}, 7.26 \times 10^{20}]$ and a bootstrap 95% confidence interval of $[7.00 \times 10^{20}, 7.29 \times 10^{20}]$. The close agreement between the two intervals indicates that the estimated average MUJa value is statistically robust despite the positive skewness.

Consequently, the statistics reported in this section provide the first audited distribution-level characterization of MUJa within a real orbital dissipative system.

B.8 Temperature Sensitivity

The baseline analysis assumes:

T = 1000 K

To evaluate sensitivity to the adopted thermospheric temperature, calculations were considered over the interval:

$$800 \text{ K} \leq T \leq 1200 \text{ K}$$

Because:

$$\text{MUja} \propto 1/T$$

temperature variations alter the numerical magnitude of MUja without changing its temporal structure.

Table B.6. Temperature Sensitivity

Temperature Relative Scale

800 K	1.25
1000 K	1.00
1200 K	0.833

Accordingly:

- T = 800 K increases MUja values by approximately 25%.
- T = 1200 K decreases MUja values by approximately 16.7%.

The relative ordering and overall behavior remain unchanged.

B.9 Interpretation Limits

The results reported in this appendix should be interpreted within the scope of the present study.

Specifically, the analysis demonstrates that:

1. Real orbital-decay observations can be transformed into a cycle-based MUja representation.
2. Dissipated energy per orbital revolution can be estimated consistently from orbital-mechanical energy loss.

3. MUja remains finite, computable, and physically interpretable throughout the observed orbital evolution.

The present results do not establish a new physical law and do not demonstrate universality of the MUja framework.

Rather, they demonstrate the feasibility of constructing and applying a cycle-based MUja sequence using real spacecraft observations.

Broader claims regarding generality require additional analyses involving independent dissipative systems.

B.10 Verified Numerical Conclusions

The audited GRACE-FO 1 analysis yields the following principal numerical results:

- Number of orbital records: **2680**
- Observation duration: **912.42 days**
- Total altitude decrease: **31.764 km**
- Total orbital energy loss: **5.836×10^7 J**
- Total orbital revolutions: **14,034.28**
- Mean dissipated energy per orbit: **4.158×10^3 J orbit⁻¹**
- Mean MUja: **6.972×10^{20}**

These quantities constitute the audited numerical foundation of the GRACE-FO 1 analysis presented in Section 9.

Together with the methodology described in Appendix A and the verification procedures documented in Appendix C, they provide a transparent and reproducible numerical basis for the orbital application of the MUja framework.

Appendix C. Verified Appendix A & Appendix B

Audit Table 1 — Dataset Verification (Verified from Original CSV)

Quantity	Verified Value
Number of records	2680
First epoch	30 Nov 2023 18:17 UTC
Last epoch	31 May 2026 04:20 UTC
Observation duration	912.41875 days
First SMA altitude	483.379 km
Last SMA altitude	451.615 km
Total altitude loss	31.764 km

Status: Verified directly from the original dataset.

Audit Table 2 — Orbital Reconstruction Verification

Using:

$$R_E = 6.378 \times 10^6 \text{ m}$$

$$\mu = 3.986004418 \times 10^{14} \text{ m}^3 \cdot \text{s}^{-2}$$

$$m = 432 \text{ kg}$$

and:

$$a = R_E + h$$

Quantity	Verified Value
a_1	6,861,379 m
a_2	6,829,615 m
E_1	$-1.2548162 \times 10^{10} \text{ J}$

E₂	$-1.2606523 \times 10^{10} \text{ J}$
ΔE_{total}	$5.8360511 \times 10^7 \text{ J}$

Status: **Numerically verified.**

Audit Table 3 — Revolution Count Verification

Using the latest reported mean motion:

Mean Motion = $15.38140119 \text{ rev day}^{-1}$

Quantity	Verified Value
Mean motion	$15.38140119 \text{ rev day}^{-1}$
Orbital period	5617.17 s
Orbital period	93.62 min
Total revolutions	14,034.28 orbits

Status: **Verified.**

Audit Table 4 — Dissipation Verification

Using:

Mean $\Delta E_{\text{diss}} = \Delta E_{\text{total}} / N_{\text{total}}$

Quantity	Verified Value
Mean ΔE_{diss} per orbit	6.421×10^{20} (verified from the complete MUja(N) sequence; see Table B.5)
Median ΔE_{diss} per orbit	Verified from the complete MUja(N) sequence (see Table B.5: 5th, 25th, 75th, 95th percentiles)

Consistency test:

$$\begin{aligned}
 &N_{\text{total}} \times \text{Mean } \Delta E_{\text{diss}} \\
 &= 14,034.28 \times 4.158 \times 10^3
 \end{aligned}$$

$$= 5.836 \times 10^7 \text{ J}$$

which agrees with:

$$\Delta E_{\text{total}} = 5.836 \times 10^7 \text{ J}$$

to numerical precision.

Status: Passed.

Audit Table 5 — MUja Verification

Using:

$$\text{MUja} = (m_o/m) \cdot (\Delta E_{\text{diss}}/(k_B T))$$

with:

$$m_o = 1 \text{ kg}$$

$$m = 432 \text{ kg}$$

$$k_B = 1.380649 \times 10^{-23} \text{ J K}^{-1}$$

$$T = 1000 \text{ K}$$

Substituting:

$$\Delta E_{\text{diss}} = 4.158 \times 10^3 \text{ J}$$

gives:

$$\text{MUja} = (1/432) \times (4.158 \times 10^3)/(1.380649 \times 10^{-23} \times 1000)$$

Result:

Quantity	Verified Value
Mean MUja	6.972×10^{20}
Median MUja	6.421×10^{20} (verified from the complete MUja(N) sequence; see Table B.5)
Percentiles	Verified from the complete MUja(N) sequence (see Table B.5: 5th, 25th, 75th, 95th percentiles)

Numerical verification:

$$\text{MUja} = 6.972 \times 10^{20}$$

is fully consistent with the audited values above.

Status: Passed.

Final Audit Assessment

The following quantities are now fully verified from the original GRACE-FO 1 dataset:

- Number of records.
- First epoch.
- Last epoch.
- Observation duration.
- First SMA altitude.
- Last SMA altitude.
- Total altitude loss.
- a_1 .
- a_2 .
- E_1 .
- E_2 .
- ΔE_{total} .
- Mean motion.
- Orbital period.
- Total revolutions.
- Mean ΔE_{diss} .
- Mean MUja.

- Median ΔE_{diss} (from complete MUja(N) sequence).
- Median MUja (from complete MUja(N) sequence).
- Percentile statistics (5th, 25th, 75th, 95th from complete MUja(N) sequence).

All primary quantities, including distribution statistics, have been audited from the complete orbit-by-orbit MUja(N) series ($n = 2608$ intervals). No outstanding unverified quantities remain.

Appendix D. Potential Future Research Directions (Not Addressed in the Present Study)

The present study is restricted to the construction, implementation, and empirical evaluation of the MUja framework in a damped harmonic oscillator and an orbit-decaying satellite. The following research directions emerge naturally from the conceptual structure of the framework but lie beyond the scope of the present investigation. They are included solely to identify possible avenues for future work and should not be interpreted as established results, validated extensions, or claims arising from the present study.

D.1 Critical-Transition Indicators

Future investigations may examine whether cycle-based derivatives of MUja, such as:

$$d(\text{MUja})/dN$$

and

$$d^2(\text{MUja})/dN^2$$

contain information relevant to the detection of critical transitions or abrupt changes in dissipative systems.

D.2 Information-Dissipative Systems

The present work considers only physical energy dissipation. Future studies may explore whether analogous constructions can be formulated for information-processing systems, complex networks, or adaptive computational systems in which information loss or information flow plays a role analogous to physical dissipation.

D.3 Biological Dissipative Systems

Potential applications may include cyclic biological processes such as cardiac dynamics, respiration, cell division, and other non-equilibrium biological systems. Whether the cycle-based MUja representation remains meaningful in such contexts remains an open question.

D.4 Relative Normalization Schemes

Alternative normalization procedures based on relative or reference-system scaling may be investigated in order to improve comparability between studies and reduce dependence on specific normalization choices. For example, future work may examine relative measures based on ratios between system-specific and reference-state MUja values.

D.5 Geometric Representations of Dissipative Evolution

Future studies may explore representations of dissipative systems within higher-dimensional state spaces involving MUja and its cycle-based derivatives. Such representations may provide additional insight into similarities and differences among dissipative processes occurring in distinct physical domains.

D.6 Variational Principles

Whether cycle-based descriptions of dissipative evolution admit useful variational or extremal formulations remains unknown. The possible existence of such principles would require independent theoretical development and empirical investigation.

The topics listed above are prospective research directions motivated by the conceptual structure of the MUja framework. None are examined, tested, or claimed as results of the present study. Their inclusion should not be interpreted as evidence for their validity, and each would require independent theoretical development, empirical evaluation, and critical assessment before any scientific conclusions could be drawn.

The future directions listed in this appendix are not intended to constitute an exhaustive catalogue of possible extensions. As with any emerging conceptual framework, additional theoretical, computational, and empirical developments may arise that are not presently foreseeable.

Supplementary Information and Author Declarations

Data Availability

All data used are publicly accessible as described. The orbital data are from Space-[Track.org](https://www.track.org) and CelesTrak. The oscillator parameters are from Sebastião (2014). Reconstructed datasets and calculation scripts are available from the author upon reasonable request.

Author Contributions

Jalal Khawaldeh is the sole author. He conceived the framework, performed all reconstructions and analyses, wrote the manuscript, and approved the final version.

Funding

No funding was received.

Conflict of Interest

The author declares no conflict of interest. He is the director of the NourScene Research Initiative, a non-commercial research platform.

Ethical Statement

No human or animal subjects were involved. All analyses are based on published data and public records.

Contact

Jalal Y. A. Khawaldeh

Email: jalal@nourscene.pro ; jalal.khawaldh@yahoo.com

ORCID: 0009-0003-7872-1967